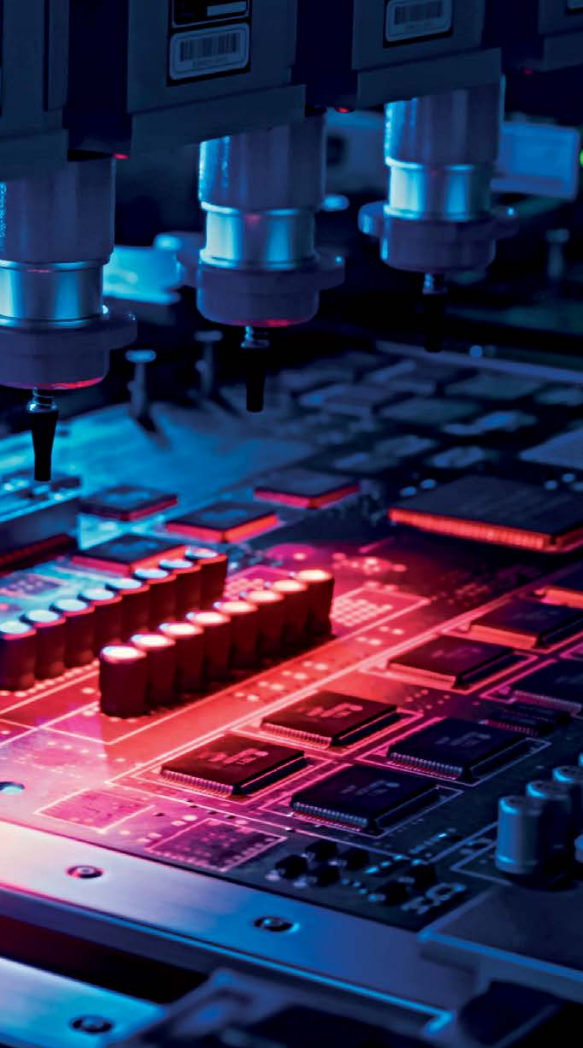


ASSEMBLY AND PACKAGING

60%

○ Around 10 OSAT houses set up via ISM1.0 provide the nation substantial capacity of 60 million product output each day when fully operational, thus reducing the dependence



PRODUCT TEST AND RELIABILITY

55%

○ Import dependence across test equipment and reliability chambers. Skilled workforce exists in India

THE HARDWARE MONOPOLY

A modern semiconductor fab is, in many ways, a collection of the most complex machines ever built. While India is investing billions of dollars to establish domestic manufacturing, it cannot yet make the equipment required to produce chips.

Photolithography systems come primarily from ASML in the Netherlands. Etch and deposition tools are supplied by American giants such as Lam Research and Applied Materials, while critical track systems are manufactured by Japan's Tokyo Electron.

India will continue to rely on global supply chains for semiconductor manufacturing equipment in the foreseeable future. Case in point is Tata Electronics, which is building India's first commercial silicon fab in Dholera, Gujarat. "Tata Electronics is partnering with PSMC in Taiwan. Initially, Tata Electronics will be using PSMC's global supply chain," says Raja Manickam, Founder & CEO, iVP Semi.

This dependence reflects the nature of the semiconductor equipment industry, which is not a commodity market. These machines require decades of research and development, deep engineering expertise and billions of dollars to be built. The decision to establish manufactur-

ing facilities is also driven by the presence of a sufficiently large and sustainable customer base.

For instance, Applied Materials currently manufactures across a globally distributed footprint, with major manufacturing units in regions such as the US, Europe and Asia. But expanding manufacturing into new geographies is closely tied to broader ecosystem readiness.

"Semiconductor equipment capability is inherently a long-cycle effort that requires sustained focus, strong partnerships and ecosystem development. Our model reflects this, with R&D, engineering and supplier ecosystems are built close to talent hubs and customer locations. Manufacturing is anchored in regions with mature supply chain ecosystems," says Avi Avula, President, Applied Materials India.

While Applied Materials' current focus in India is on product development, engineering and R&D, the company operates India's only facility capable of processing 300-mm wafers. Its India Validation Center in Bengaluru serves as a hub for equipment design, simulation, testing, engineering and process validation; and an advanced Class 1K cleanroom supports inspection, diagnostics, prototyping and component qualification.